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D025711(025)

B. Tech. (Seventh Semester) Examination, Nov.-Dec. 2022

(AIC Scheme)

(Electrical & Electronics Engg. Branch)

POWER SYSTEM PROTECTION

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *Attempt all questions. Part (a) of each question is compulsory and each carry 4 marks. Part (b), (c) and (d) any* ***two*** *parts solve each carry 8 marks.*

Unit-I

1. (a) Explain what you understand by primary and back-up protection. What is the role of back-up protection?

(b) Explain with sketches the construction and operation of IDMT relay with its characteristic.

(c) Give the constructional features, principle of working of directional over current relay. Draw its characterisitcs.

(d) What are the essential features required for directional relay? Explain the principle of single phase directional relay showing its characteristics?

Unit-II

2. (a) State the various types of faults that occur in an alternator.

(b) Describe the magnetizing in rush protection scheme for power transformers.

(c) A 3-phase, 66/11 kV star-delta connection transformer is protected by Merz Price system. If CT's on LT side have a ratio of 420/5. Determine ratio of CT's on HT side.

(d) Discuss the protections scheme used for intertum faults in stator of alternator.

Unit-III

3. (a) What is meant by frame leakage protection?

(b) What is carrier current protection? With neat sketches, discuss the phase comparison schemes of carrier current protection.

(c) Give various schemes of protection of feeders and compare their performance.

(d) Discuss the effect of power swings on distance relays in transmission lines protection.

Unit-IV

4. (a) What are the merits and demerits of static relay over electromagnetic relay?

(b) Describe the operation of hall effect relay used as a phase comparator.

(c) Explain quadrilateral characteristics relay by using cosine phase comparator.

(d) Discuss how an amplitude comparator can be converted to a phase comparator and vice versa?

Unit-V

5. (a) What is meant of current chopping?

(b) Explain the construction and principle operation with application of Air Blast Circuit Breaker.

(c) Explain the terms recovery voltage and factor affecting recovery voltage.

(d) Draw SF6 circuit breaker, enumerating its various components. What is meant by making and breaking of a circuit with reference to circuit breaker operation?